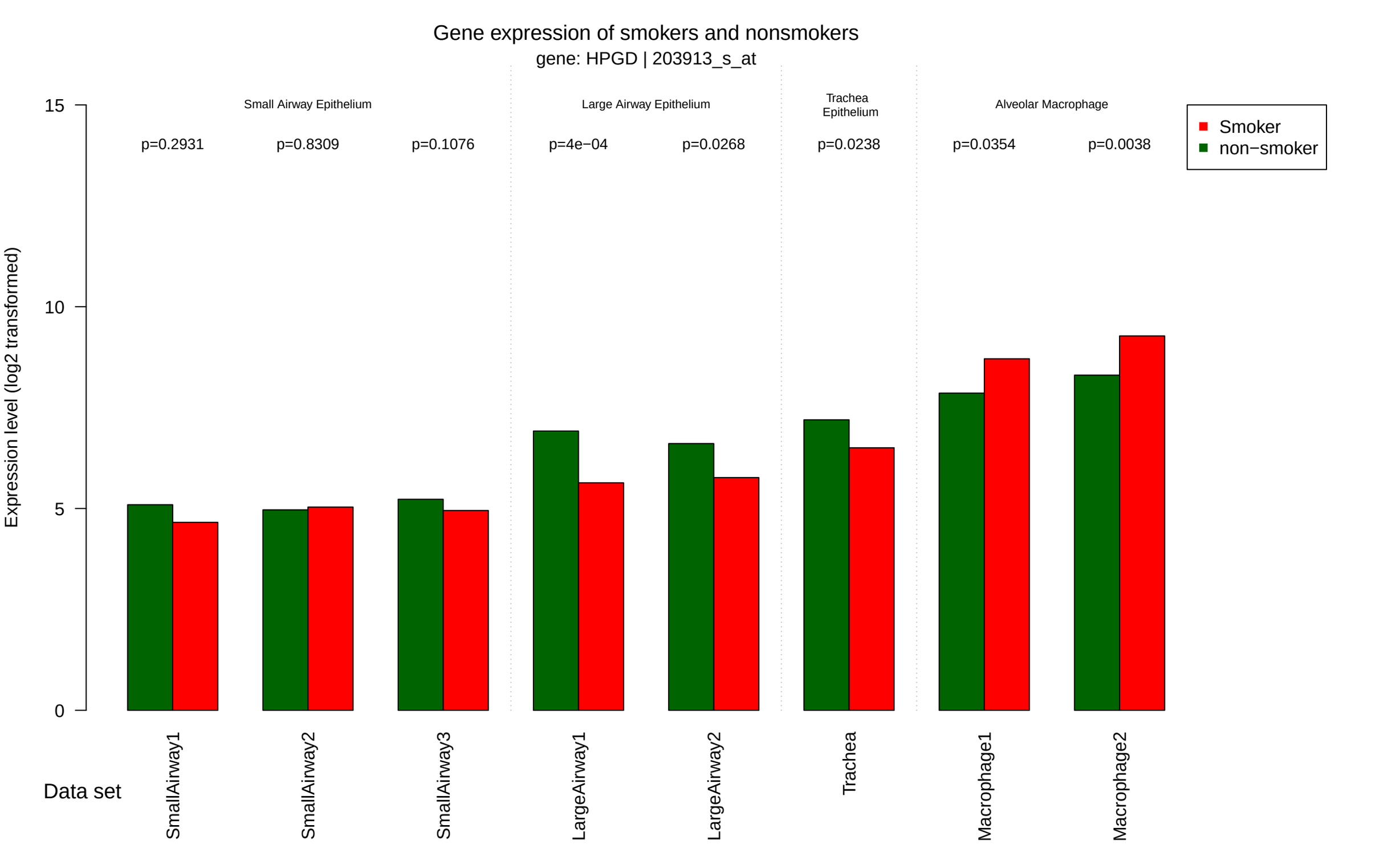
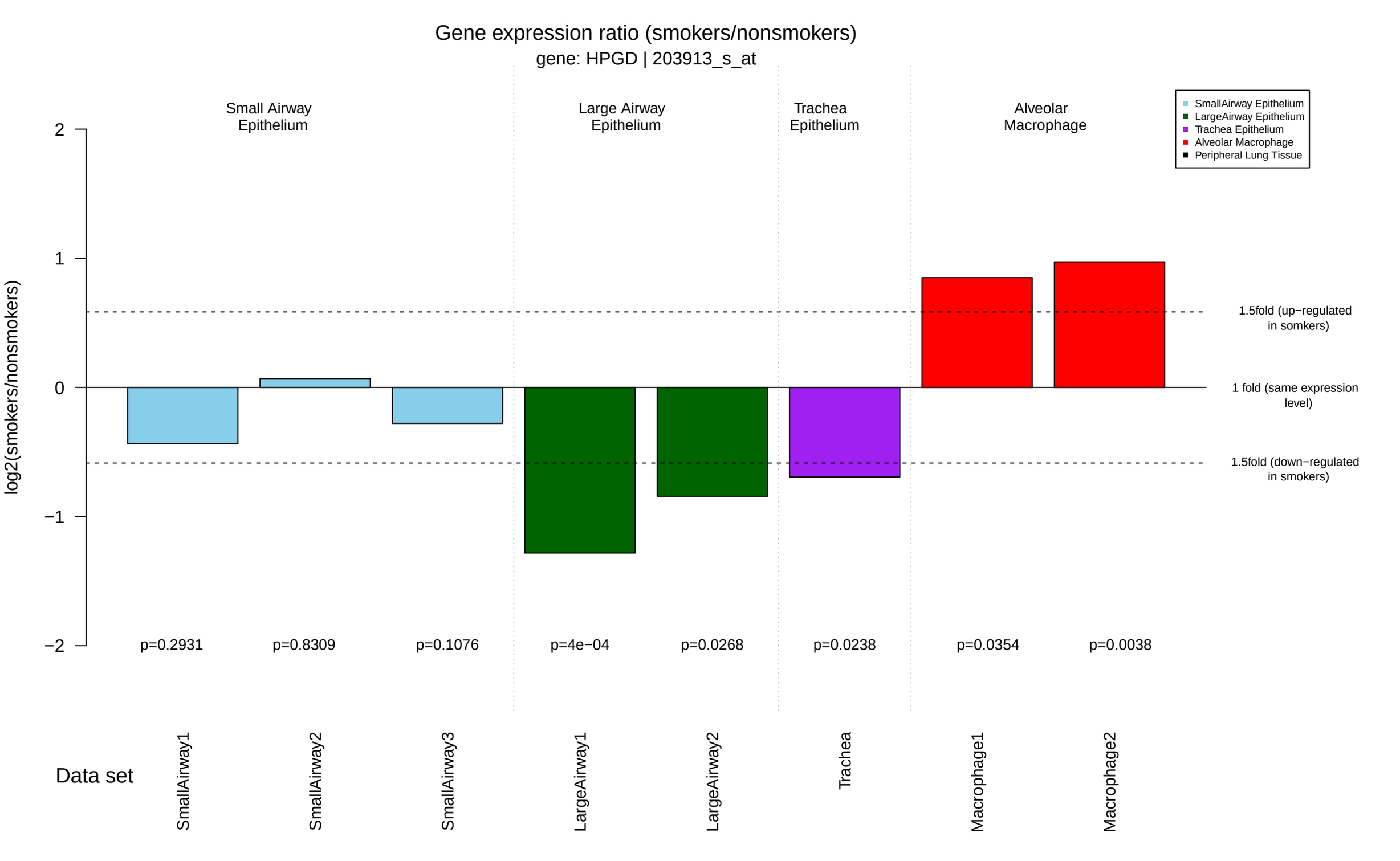
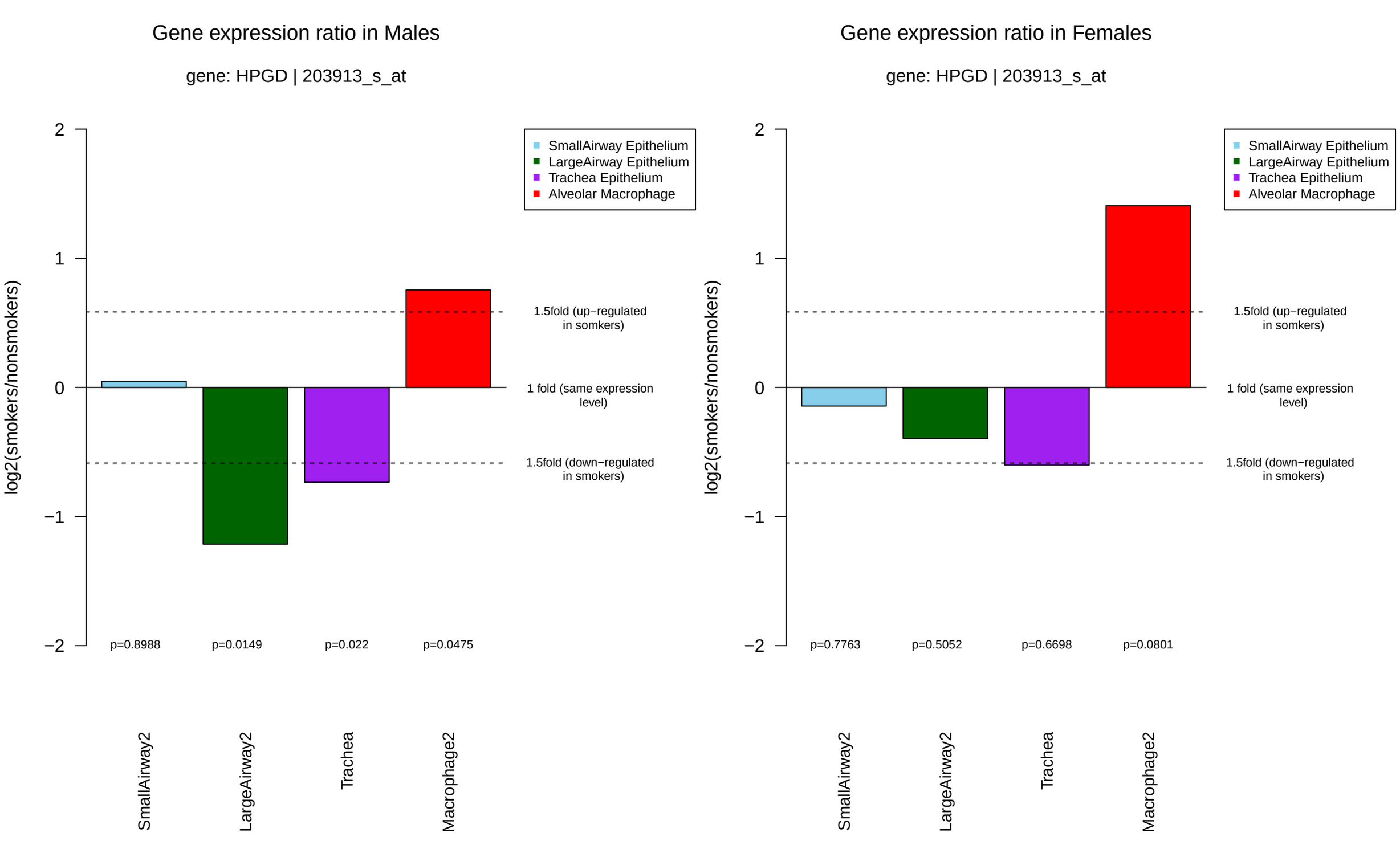
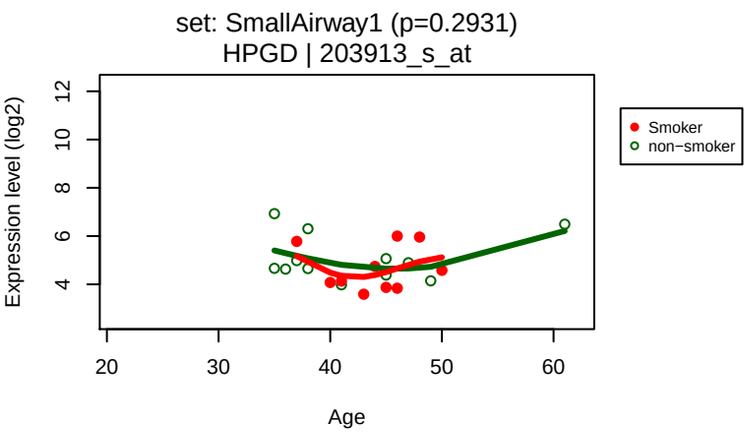
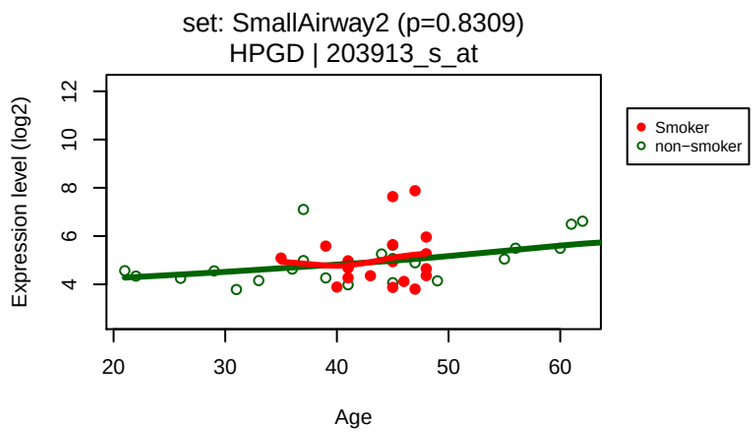
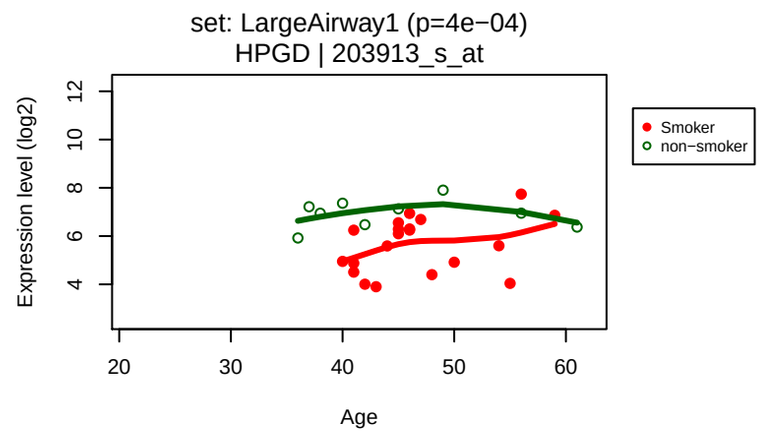

Gene expression of smokers and nonsmokers
gene: HPGD | 203913_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2931
p=0.8309
p=0.1076
p=4e−04
p=0.0268
p=0.0238
p=0.0354
p=0.0038
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

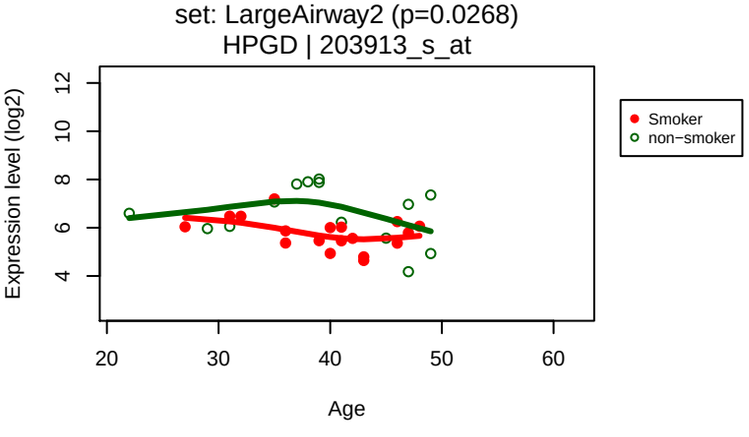

Gene expression ratio (smokers/nonsmokers)
gene: HPGD | 203913_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2931
p=0.8309
p=0.1076
p=4e−04
p=0.0268
p=0.0238
p=0.0354
p=0.0038
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

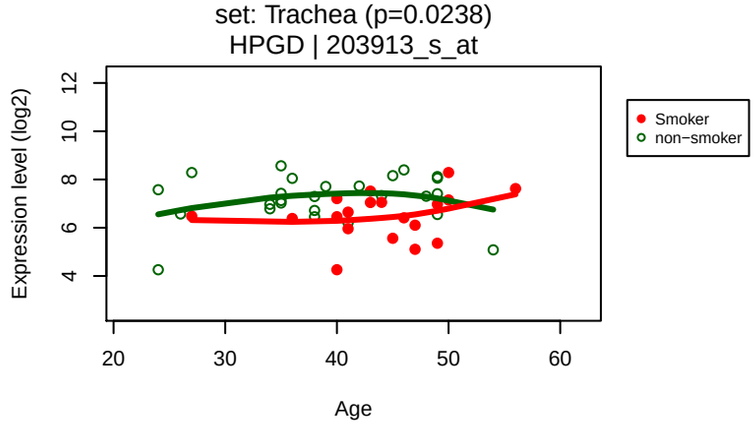

Gene expression ratio in Males
gene: HPGD | 203913_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8988
p=0.0149
p=0.022
p=0.0475
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPGD | 203913_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7763
p=0.5052
p=0.6698
p=0.0801
SmallAirway2
LargeAirway2
Trachea
Macrophage2

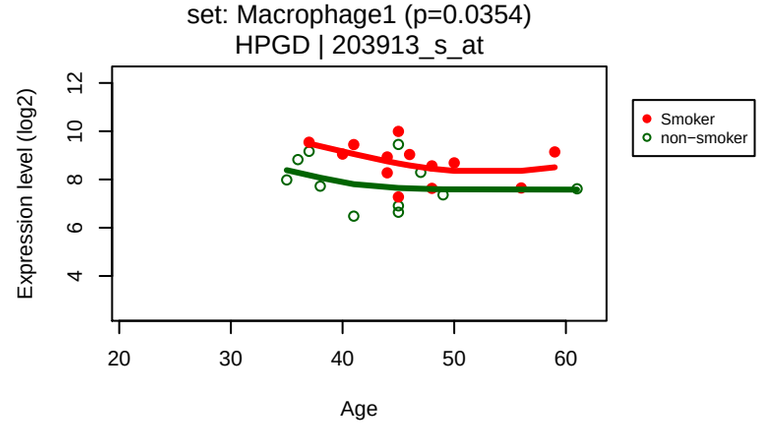

set: SmallAirway1 (p=0.2931)
HPGD | 203913_s_at
Expression level (log2)
Age
Smoker
non−smoker

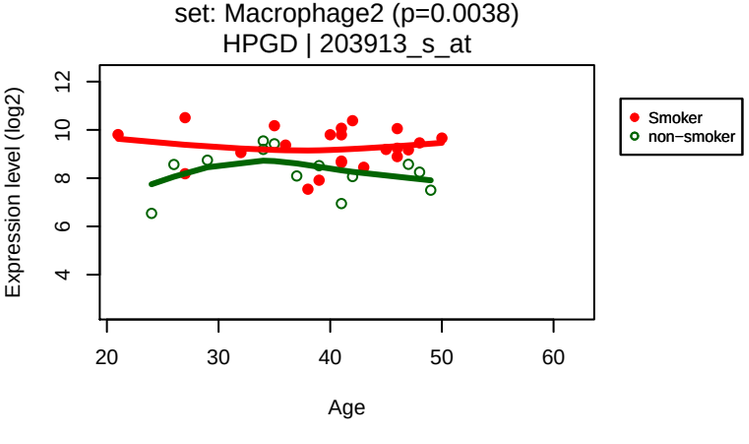

set: SmallAirway2 (p=0.8309)
HPGD | 203913_s_at
Expression level (log2)
Age
Smoker
non−smoker

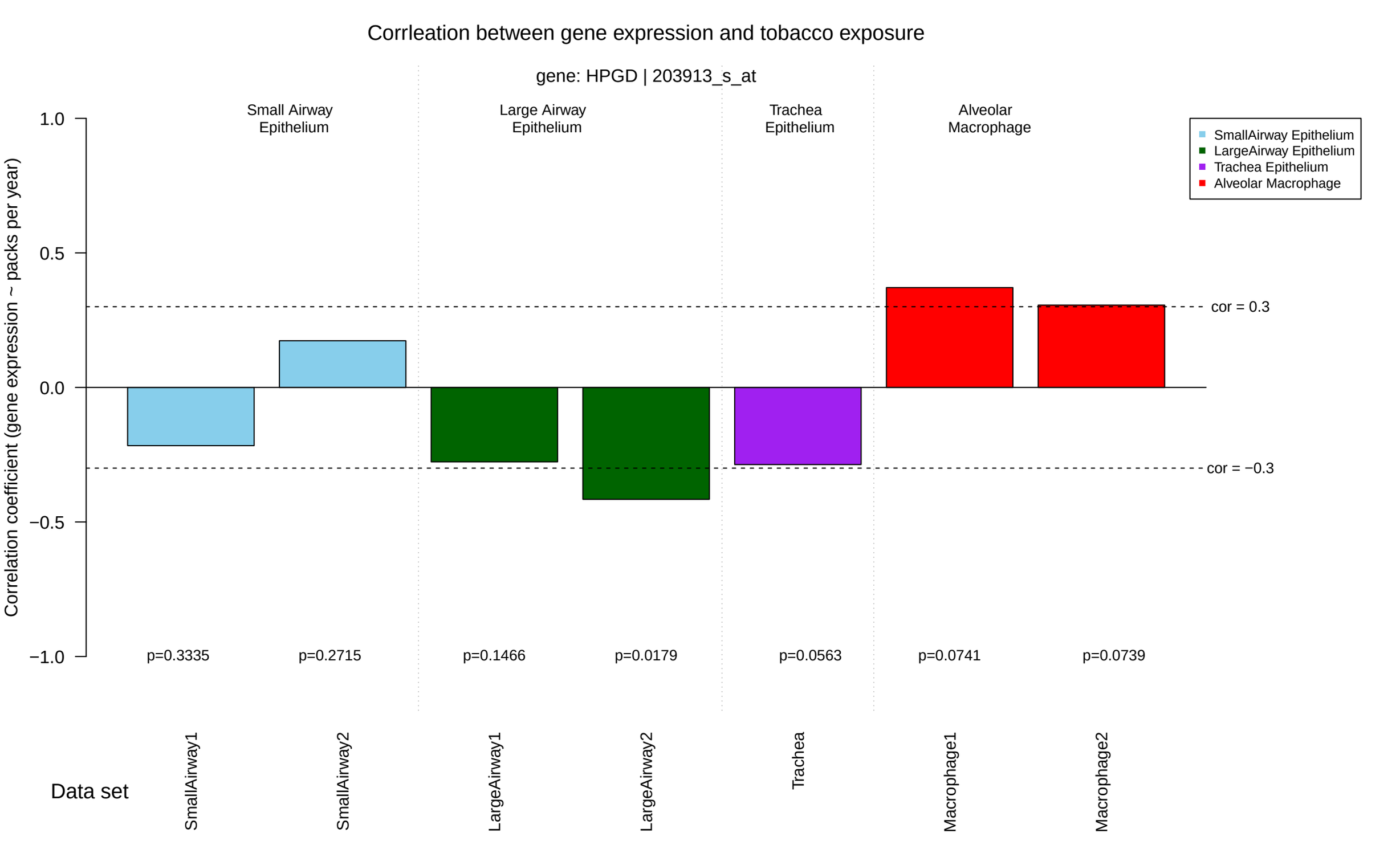

set: LargeAirway1 (p=4e−04)
HPGD | 203913_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0268)
HPGD | 203913_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0238)
HPGD | 203913_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0354)
HPGD | 203913_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0038)
HPGD | 203913_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPGD | 203913_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3335
p=0.2715
p=0.1466
p=0.0179
p=0.0563
p=0.0741
p=0.0739
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2